OSS Audit Project Report

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Course: Open Source Software Audit

Software Selected: Apache HTTP Server

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1. Introduction

Open Source Software (OSS) plays a crucial role in modern computing by providing free, flexible, and community-driven solutions. Apache HTTP Server is one of the most widely used open-source web servers in the world.

The objective of this project is to audit and analyze Apache HTTP Server using shell scripting techniques. This includes gathering system information, checking installation status, analyzing disk usage, and evaluating log files.

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2. Objectives

- To understand the basics of open-source software auditing
- To learn shell scripting for system-level tasks
- To analyze system configuration and performance

- To check the presence of Apache HTTP Server

- To work with Linux commands for real-time auditing

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3. Methodology

The project was implemented using Bash scripting on a Linux-based environment. Each script performs a specific auditing task. The scripts were developed, tested, and stored in a GitHub repository for version control and submission.

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4. Scripts Description

Script 1: System Identity Report

This script collects system-level information such as:

- Operating System distribution

- Kernel version

- Logged-in user

- System uptime

- Current date and time

This helps in understanding the environment where Apache HTTP Server is running.

Script 2: Package Check

This script checks whether Apache HTTP Server is installed in the system using package management commands.

- If installed → displays confirmation
- If not installed → displays a warning message

Script 3: Disk Usage Analysis

This script checks disk usage of important directories:

- /etc (configuration files)
- /var (logs and server data)
- /home (user data)

This helps in identifying storage usage and potential issues.

Script 4: Log Analyzer

This script analyzes system or application log files.

- It searches for error entries
- Counts total number of errors
- Helps in detecting issues in Apache server logs

Script 5: Manifesto Script

This is an interactive script that:

- Takes user input
- Displays a message supporting open-source tools

It highlights the importance of open-source contributions.

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5. Tools and Technologies Used

- GitHub (for version control and submission)
- GitHub Desktop (for repository management)
- Visual Studio Code (for coding)
- Bash Shell (for scripting)
- Linux Commands (uname, uptime, grep, etc.)

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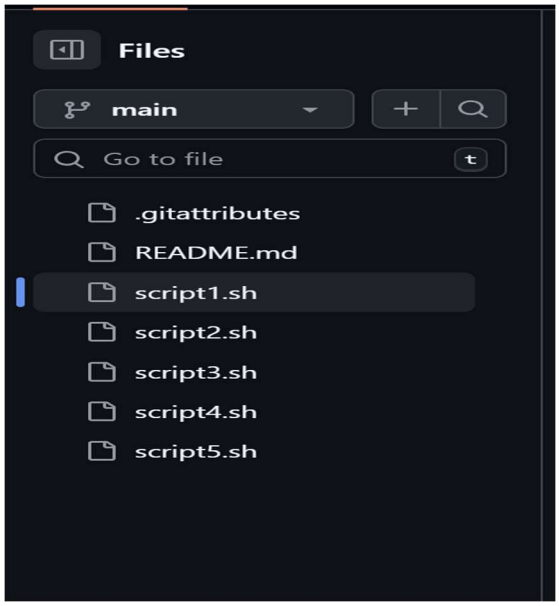
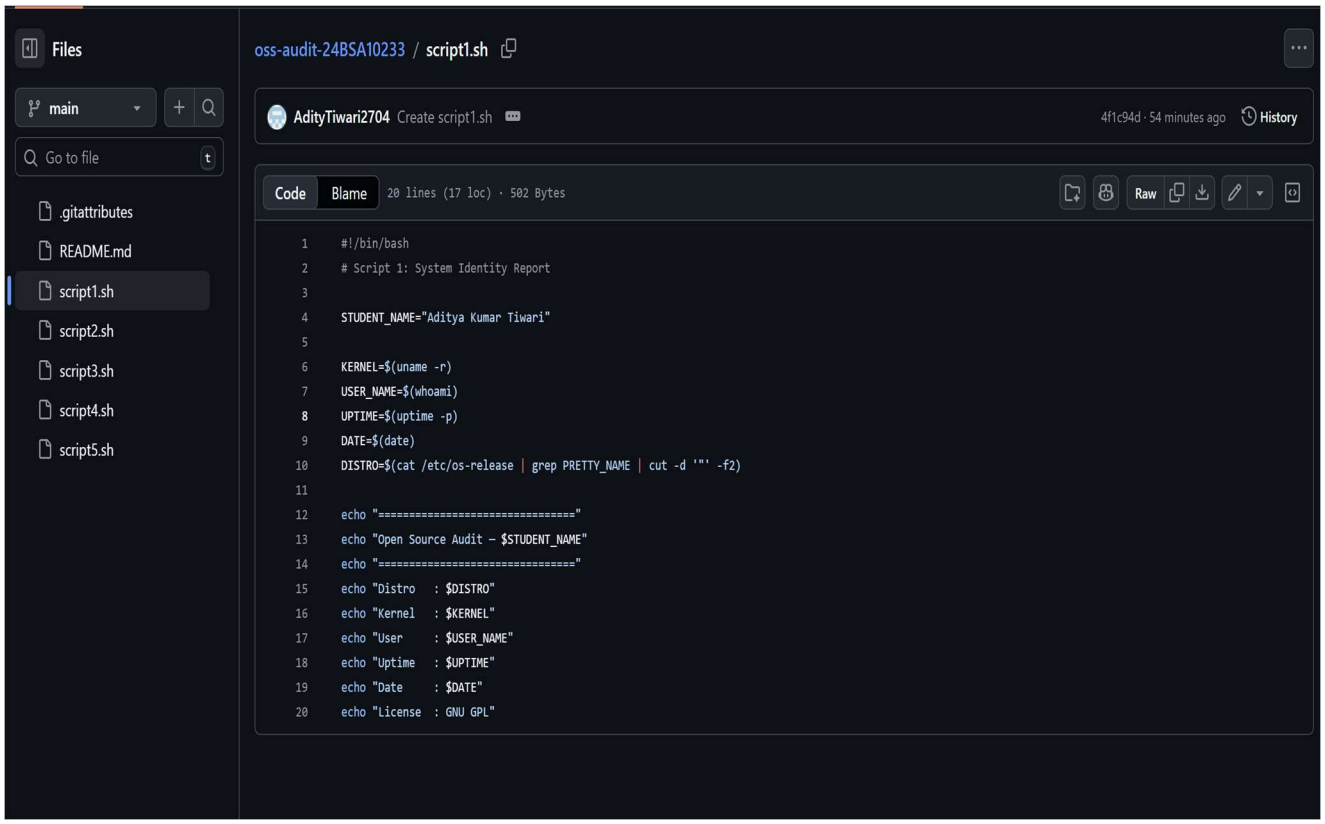
6. Screenshots

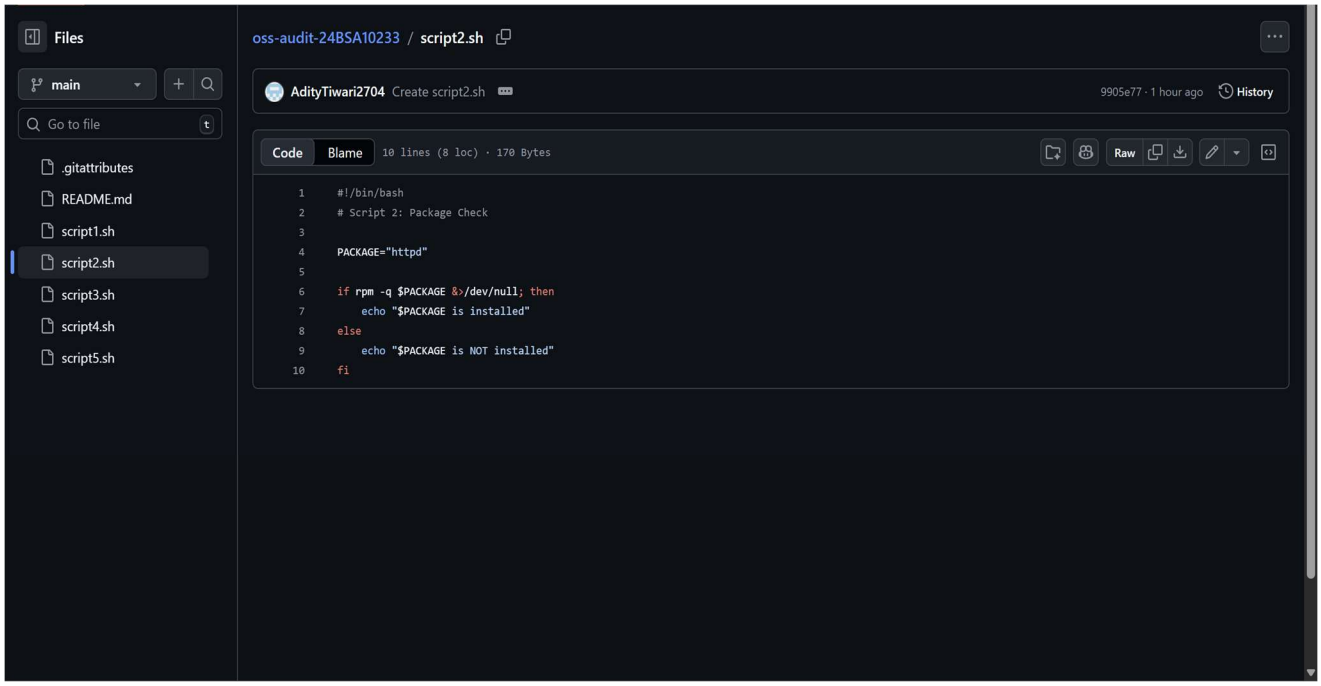
1. GitHub repository showing all files



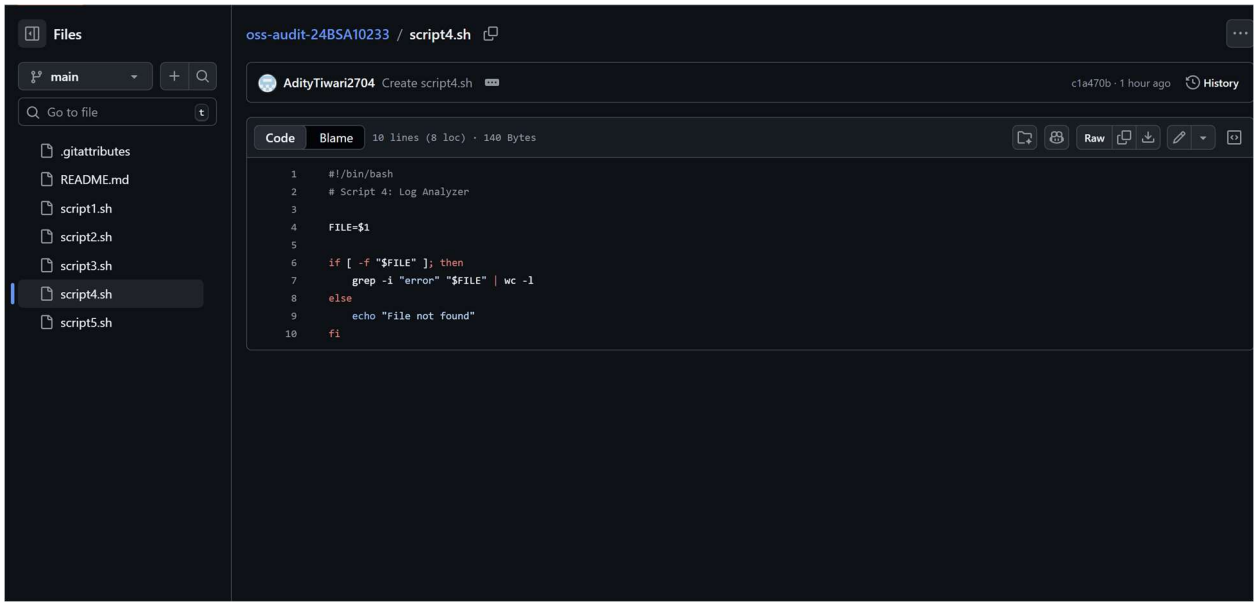
2. script1.sh code



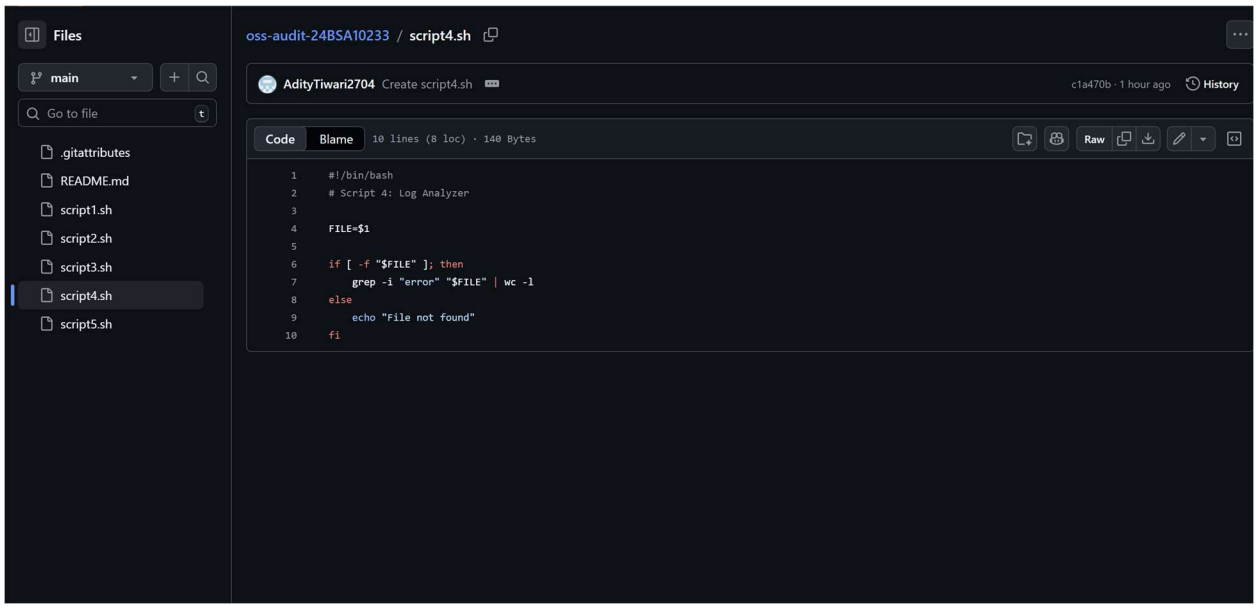
3. script2.sh code



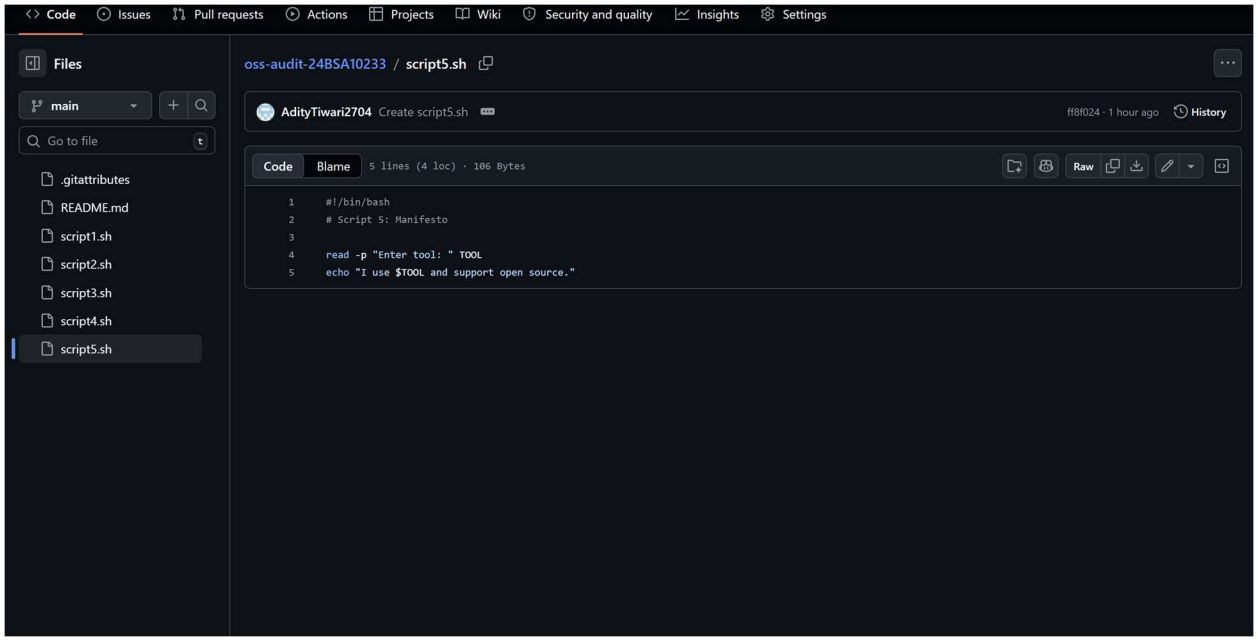
4. script3.sh code



5. script4.sh code



6. script5.sh code



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7. Results and Observations

- Successfully retrieved system information using Bash scripts
- Verified Apache HTTP Server installation status
- Analyzed disk usage effectively
- Extracted meaningful data from log files
- Improved understanding of Linux commands and scripting

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8. Conclusion

This project provided hands-on experience in open-source auditing and shell scripting. It helped in understanding how system-level analysis can be automated using simple scripts. Apache HTTP Server was successfully analyzed, and key insights were obtained.

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9. GitHub Repository Link

https://github.com/AdityTiwari2704/oss-audit-24BSA10233